

YASH RAJ PANDEY

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Summary

AI systems engineer. I build local-first LLM infrastructure, agent platforms, and the evaluation systems that decide what ships. I also reproduce new model research, publish what breaks, and fix it upstream: 65 merged pull requests across 28 projects, including llama.cpp, MLX, RAGFlow and LiteLLM.

Experience

University of Florida

Gainesville, FL

AI Agents Architect

Apr 2026 – Present

- Architected and now run a production AI agent platform: retrieval over 10,000+ technical documents, tool-calling workflows, and natural-language access to research data.
- Replaced about 20 colliding rule-based routers with one deterministic intent arbiter, removing a class of wrong-tool failures the model could not detect.
- Built the release gate that decides what ships: 26 task-specific evaluations and a 24-case regression suite across 14 candidate models. It caught hallucination failures before deployment.
- Led the migration of generation onto university-operated GPUs for a 120B-parameter open-weight MoE. Retrieval and reranking stay local, where a cross-encoder measured better.

Lead Software Engineer

Oct 2025 – Apr 2026

- Led Blue Omics, a Django, React and PostgreSQL genomics platform: system of record for 30+ researchers across 5 labs, scaled to 5.38M+ records.
- Cut end-to-end R Markdown reporting from 2–3 hours to 15–20 minutes, and core lookup latency to 23–29 ms.
- Shipped 7 ingestion and validation pipelines, audited 1.4M+ cells with zero mismatches, and grew the team from one to three.

Software Engineer

Mar 2025 – Oct 2025

- Designed the original PostgreSQL schemas, Python ingestion and validation logic; cut frontend load time from 8s to 3s.

Selected Projects

Open Source / 65 merged pull requests across 28 projects

github.com/devYRPauli

- **llama.cpp** (126k stars), **RAGFlow** (89k), **mem0** (64k), **LiteLLM** (57k) and **MLX** (28k): correctness fixes in inference kernels, document parsers, vector retrieval and per-request billing. Each starts with a failure I reproduced and ends with a regression test.

willitcall / A compatibility matrix for tool calling on local models

github.com/devYRPauli/willitcall

- Rust CLI that runs one conformance suite against any OpenAI-compatible endpoint. Publishes 32 rows across 3 serving stacks. Nobody had published the matrix.

Looma / Local-first memory for coding agents

github.com/devYRPauli/looma

- Zero-dependency Python CLI turning Claude Code, Codex and Cursor transcripts into resumable context. F1 0.86, 134 tests, published on PyPI.

Podium / Verified delegation for coding agents

github.com/devYRPauli/podium

- A runner executes the acceptance check and a shell exit code settles it, so a bot cannot report its own success.

Independent Evaluations / TurboQuant KV-cache compression and Google TabFM

yashrajpandey.com/writing

- Restored TurboQuant needle retrieval from 0% to 100% at 16K on a 16 GB M1 Pro by root-causing five bugs. Landed a multi-GPU crash fix in google-research/tabfm.

Technical Skills

AI systems: LLM inference and evaluation, RAG, agent runtimes, tool calling, vLLM, llama.cpp, MLX, quantization, vector search, reranking, MCP

Languages: Python, TypeScript, JavaScript, SQL, C/C++, Rust, Bash

Frameworks and data: Django, FastAPI, React, Next.js, PostgreSQL, DuckDB, SQLite, Redis, Qdrant, Supabase

Infrastructure: Docker, Linux, GPU and self-hosted inference, GCP, Vercel, Git, GitHub Actions

Education

University of Florida – M.S. Computer & Information Science & Engineering (GPA 3.8)

Aug 2023 – Dec 2024

Jaypee University of Engineering and Technology – B.Tech. Computer Science

Jul 2019 – May 2023